

University of Freiburg
Research Training Group 3186

Dear Members of the Selection Committee,

My interest in theoretical physics has been driven by a desire to understand the fundamental structure of nature, which has naturally led me towards quantum field theory and particle physics. **I am particularly fascinated by the role of symmetries in shaping physical laws**, including how phenomena such as spontaneous symmetry breaking and anomalies play a central role in determining particle behaviour. I wish to deepen my understanding of these principles and ultimately piece the clues together to answer the question: **what symmetries govern physics beyond the Standard Model?**

My research experience spans fundamental particle physics, cosmology, and computational methods. Currently, I am working on my master's thesis, which focuses on gaining a deeper understanding of fundamental interactions and **beyond-the-Standard-Model physics**. In my research, I explore grand unified theories (GUTs). Rather than proposing a specific gauge group, **I study the landscape of GUTs, and aim to extend this approach to include scalars**, by exploring how their representations affect model viability. This work is currently being prepared for publication. In parallel, I am involved in a project on infinite heat order, concerning **high-temperature symmetry non-restoration in UV-complete models**. During my master's degree, while on exchange in Canada, I have completed another project on topological anomalies in quantum field theory.

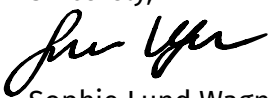
During my bachelor's degree, I worked on density profiles of dark matter, both theoretically and using machine learning techniques. This interest culminated in a bachelor's thesis in cosmology, where I examined whether inhomogeneities in the structure of the universe could explain its accelerated expansion.

For my PhD, I want to contribute to research in theoretical particle physics. I yearn to use the framework of quantum field theory to study the interplay of flavour, Higgs dynamics, and beyond-Standard-Model physics, with the hope of uncovering the deeper principles governing fundamental interactions. **In particular, I am interested in phenomenological approaches to beyond-the-Standard-Model physics**, where Higgs-sector dynamics and precision quantum field theoretical calculations serve as sensitive probes of new physics, **in close alignment with the research directions pursued by Heidi Rzehak and Stefan Dittmaier**.

I am excited to apply to the the Research Training Group at University of Freiburg due to its strong expertise in theoretical particle physics and its close connection to experimental developments. **I am particularly attracted to the research on particle phenomenology and beyond-the-Standard-Model physics**. I would therefore be especially interested in learning from and contributing to research directions pursued by researchers such as Heidi Rzehak.

Thank you for considering my application. I would be thrilled to contribute to the research activities of the Research Training Group at University of Freiburg.

Sincerely,



Sophie Lund Wagner

Sophie Lund Wagner

 Sophie Lund Wagner |  sowag21@student.sdu.dk |  +4542187737

EDUCATION

M.S.c in Physics at University of Southern Denmark, Denmark 2024 – 2026

- Thesis title: *Novel principles for grand unification theories.*
- Thesis advisor: Francesco Sannino.

Exchange semester at University of Calgary, Alberta, Canada 2025

B.S.c in Physics at University of Southern Denmark, Denmark 2021 – 2024

- Thesis title: *Inhomogeneous cosmology and cosmic backreaction.*
- Thesis advisor: Sofie Marie Koksang.

RESEARCH INTERESTS

Theoretical physics, particle physics, fundamental interactions, cosmology.

AWARDS AND SCHOLARSHIPS

Zonta Denmark: “Women in STEM, 2024”, Niels Bohr Foundation, William Demant Foundation, Knud Højgaard Foundation, Torben and Alice Frimodts Foundation, Doctor Else Mogensen Foundation

PARTICIPATION AT SCHOOLS/CONFERENCES

IDA Summer School, NOT telescope La Palma 2025

TEACHING EXPERIENCE

High School project supervisor (University of Southern Denmark, Denmark) 2024 – present

- teaching and guidance of high school students in their final project

OUTREACH

Outreach instructor (University of Southern Denmark, Denmark) 2024 – present

- coordination and developing of “physics clubs” at high schools in Southern Denmark, planning of physics-events, teaching at summer schools (supported by the *Villum Foundation*).

Student ambassador (University of Southern Denmark, Denmark) 2023 – present

PROGRAMMING SKILLS

Python, Mathematica

List of Referees

Referee 1

Francesco Sannino

Director, Quantum Theory Center ($\hbar$ QTC) and Chair of Science
Danish Institute for Advanced Study (Danish IAS)
University of Southern Denmark
Campusvej 55, DK-5230 Odense M, Denmark

Professor of Theoretical Physics
Department of Physics E. Pancini, Università di Napoli Federico II
INFN Sezione di Napoli
Scuola Superiore Meridionale
Via Cintia, 80126 Napoli, Italy

✉ sannino@qtc.sdu.dk

✉ sannino@mac.com

Referee 2

Giacomo Cacciapaglia

Senior Researcher
Laboratoire de Physique Théorique et Hautes Énergies (LPTHE), UMR 7589
Sorbonne Université & CNRS
4 place Jussieu, 75252 Paris Cedex 05, France

✉ cacciapa@lpthe.jussieu.fr

Referee 3

Sofie Marie Koksang

Associate Professor
CP3-Origins
University of Southern Denmark
Campusvej 55, DK-5230 Odense M, Denmark

✉ koksang@cp3.sdu.dk

List of Passed Subjects

Sophie Lund Wagner
Falen 12, 4. th
5000 Odense C

It is hereby confirmed that Sophie Lund Wagner, date of birth: 14.12.1999, is enrolled as a student at University of Southern Denmark.

Name of the education: **Bachelor`s Programme**

The student graduated on 27.07.2024.

The student has passed the following subjects.

The credits are shown in ECTS-point

	Marks	ECTS-grade	Credits
Bachelor`s Programme, Physics	passed		180,0000
Bachelor, Physics	passed		180,0000
Bacheloruddannelsen i fysik	passed		180,0000
Compulsory Courses	passed		140,0000
Studiestartsprøve	passed		0,0000
Studiestartsprøve	Passed		0,0000
1. årsprøve fysik 2019	passed		30,0000
Mechanics and thermodynamics	12	A	10,0000
Calculus	12	A	10,0000
Electronics	12	A	5,0000
Experimental Design and Modelling	12	A	5,0000
FT504 (F20)	passed		10,0000
Electromagnetism and Optics	10	B	10,0000
Electromagnetism and Optics	Passed		0,0000
Electromagnetism and Optics	Passed		0,0000
FT506: Førsteårsprojekt	passed		10,0000
FT506: First Year Project	12	A	9,0000
Introduktionsforløb om verdensmålene for bæredygtig udvikling	Passed		1,0000
Digital Electronics and Programming	12	A	10,0000
Quantum mechanics I	12	A	5,0000
Advanced Mechanics and Relativity Theory	7	C	10,0000
Linear algebra	12	A	5,0000
FY540/FY554+statistik	passed		10,0000
FY554 + Statistik	passed		10,0000
FY554	passed		5,0000
Numerical physics	12	A	5,0000

List of Passed Subjects

Sophie Lund Wagner, date of birth: 14.12.1999

Numerical physics	Approved		0,0000
Statistics	12	A	5,0000
Quantum mechanics II	12	A	5,0000
Quantum mechanics II	Passed		0,0000
Statistical Physics	12	A	5,0000
Electrodynamics	12	A	5,0000
(F) Elektrodynamik	Passed		0,0000
Experimental physics and semiconductors	12	A	5,0000
Experimental physics and semiconductors	Passed		0,0000
Fundamental Astronomy	12	A	10,0000
Solid state physics	12	A	5,0000
Astrophysics and Introduction to Cosmology	12	A	5,0000
Particle physics	12	A	5,0000
Partial Differential Equations and complex analysis	12	A	5,0000
Valgfagsmodul/Valgfag	passed		40,0000
Elective Module	passed		15,0000
FY803 (E18)	10		10,0000
Quantum physics	7	C	5,0000
Quantum physics	12	A	5,0000
FY555: Introduktion til Python	passed		5,0000
Introduction to Python, machine learning and data handling for the physical sciences	Approved		0,0000
Introduction to Python, machine learning and data handling for the physical sciences	Passed		5,0000
Elective courses	passed		10,0000
Individual Study Activity in Physics	Passed		5,0000
Title: Dark matter and neural networks			
Galactic dynamics and dark matter	10	B	5,0000
Bachelorprojekt i fysik	passed		15,0000
Bachelor's project in Physics	12	A	15,0000
Title: Cosmic backreaction			

Name of the education: Master's Programme

The student has not graduated.
The student has passed the following subjects.
The credits are shown in ECTS-point

	Marks	ECTS-grade	Credits
Master's Programme, Physics			60,0000
Master's Programme, Physics			60,0000
Kandidatuddannelsen i fysik			60,0000
Study start test for Physics	Passed		0,0000
FY829	passed		5,0000
Research activities in experimental physics	Passed		5,0000
Research activities in experimental physics	Passed		0,0000
Konst.fys.kurser/Valgfag	passed		55,0000

List of Passed Subjects

Sophie Lund Wagner, date of birth: 14.12.1999

Konstituerende kurser	passed		25,0000
Advanced statistical physics	12	A	10,0000
Quantum Field Theory	12	A	10,0000
General relativity and cosmology	12	A	5,0000
Valgfag	passed		30,0000
Individual Study Activity in Physics	Passed		5,0000
Title: Anomalies in Quantum Field Theory			
Observational astronomy	Passed		5,0000
FY826: Observational astronomy	Passed		0,0000
NAT817	passed		5,0000
Oral Science Communication	Passed		5,0000
Oral Science Communication	Passed		0,0000
Electives			
Transferred from University of Calgary, Canada	Passed		15,0000
Electrodynamics - Quantum optics - Graduate seminar			

Examinations and diplomas

All examinations at Danish higher education institutions are administered not only by the teacher, but also by an examiner who often is external. The examiners are responsible for assuring the same standard for all examinations and thus their quality. After completion of a full programme, the student is awarded a diploma and a transcript.

The 7-point grading scale

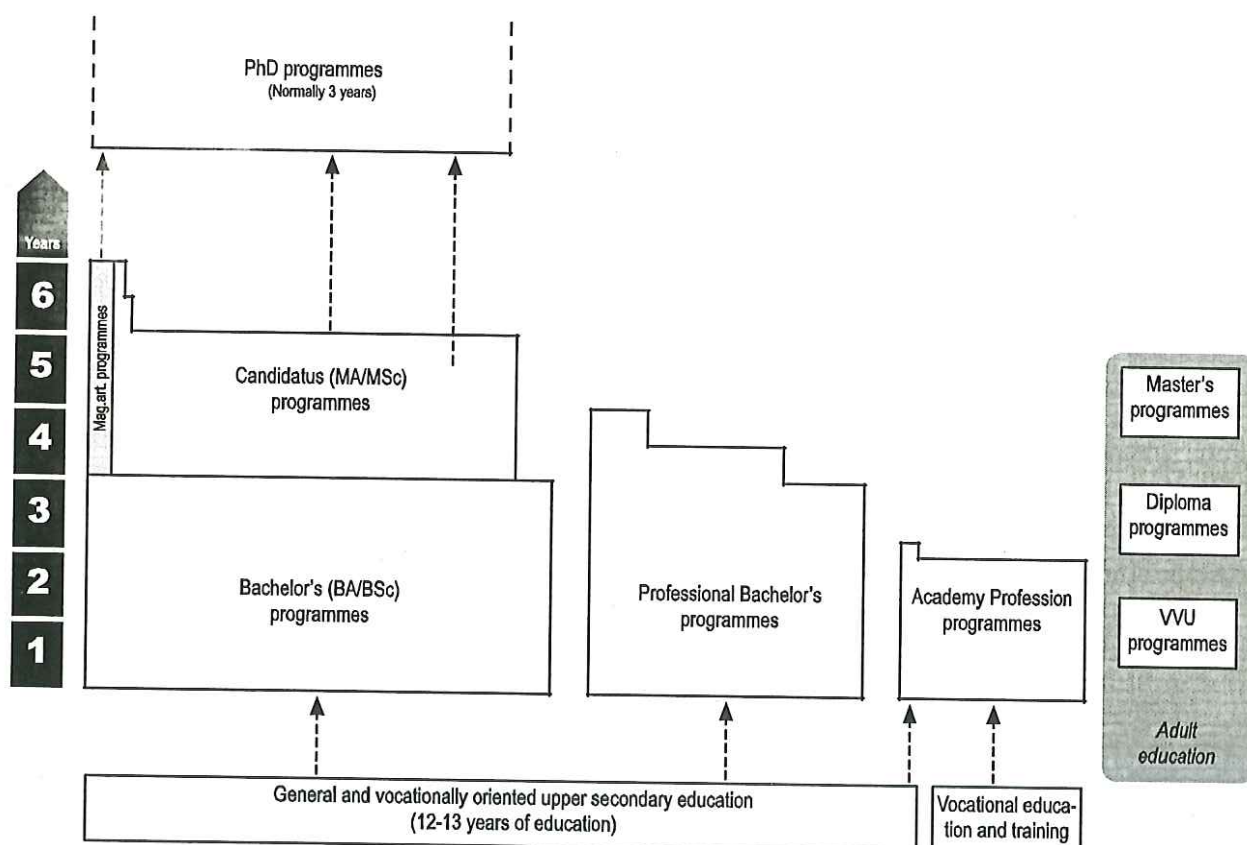
The grading system used in all state-regulated education as from August/September 2007 is the 7-point grading scale. The grading scale is compatible with the ECTS grading scale.

The 7-point grading scale		The ECTS grading scale
12	For an excellent performance displaying a high level of command of all aspects of the relevant material, with no or only a few minor weaknesses	A
10	For a very good performance displaying a high level of command of most aspects of the relevant material, with only minor weaknesses.	B
7	For a good performance displaying good command of the relevant material but also some weaknesses.	C
4	For a fair performance displaying some command of the relevant material but also some major weaknesses.	D
02	For a performance meeting only the minimum requirements for acceptance	E
00	For a performance which does not meet the minimum requirements for acceptance	Fx
-3	For a performance which is unacceptable in all respects	F

Apart from the 7-point grading scale, pass/fail assessment may also be used. 02 is the minimum grade for passing an exam.

The performance of the student must be assessed according to the academic targets set for the specific subject or course (absolute grading method). The relative grading method must not be used.

Higher Education in Denmark



Sophie Lund Wagner

CPR-nr. 141299-0214

har den
has on the

27. juli 2024

bestået
passed

Bacheloruddannelsen i fysik

The Bachelor's Programme in Physics

og har dermed ret til at betegne sig
and has thus been awarded the title of

Bachelor (BSc) i fysik

Bachelor of Science (BSc) in Physics

Syddansk Universitet, den 2. august 2024



Marianne Holmer

Dekan
Dean